

1. Object Detection Performance Comparison

Given Data

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
Model A – Viola–Jones	80	20	30
Model B – YOLO	90	15	20

(a) Precision and Recall

Formulas:

$$Precision = \frac{TP}{TP + FP}$$
$$Recall = \frac{TP}{TP + FN}$$

Model A – Viola–Jones

Precision:

$$P_A = \frac{80}{80 + 20} = \frac{80}{100} = 0.80$$
$$\boxed{Precision_A = 80\%}$$

Recall:

$$R_A = \frac{80}{80 + 30} = \frac{80}{110} = 0.7273$$
$$\boxed{Recall_A = 72.73\%}$$

Model B – YOLO

Precision:

$$P_B = \frac{90}{90 + 15} = \frac{90}{105} = 0.8571$$

$$\text{Precision}_B = 85.71\%$$

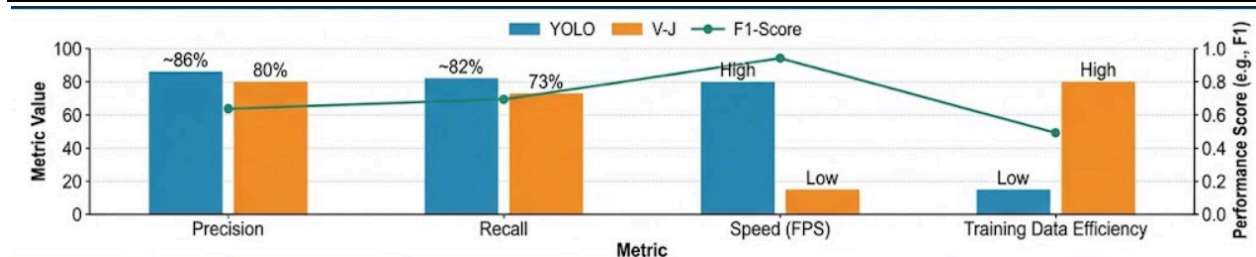
Recall:

$$R_B = \frac{90}{90 + 20} = \frac{90}{110} = 0.8182$$

$$\text{Recall}_B = 81.82\%$$

(b) Comparison

Performance Measure	Model A – Viola–Jones	Model B – YOLO
Precision	80%	85.71%
Recall	72.73%	81.82%



Therefore, YOLO is more accurate in its detections and misses fewer actual objects.

2. Motion Estimation Error Analysis

Given Data

- Method A: 2, 3, 2 pixels
- Method B: 3, 5, 4 pixels

(a) Average Displacement Error

Formula:

Average Error=Number of frames/Sum of displacement errors

Method A

$$\text{Average Error}_A = \frac{2 + 3 + 2}{3} = \frac{7}{3} = \boxed{2.33 \text{ pixels}}$$

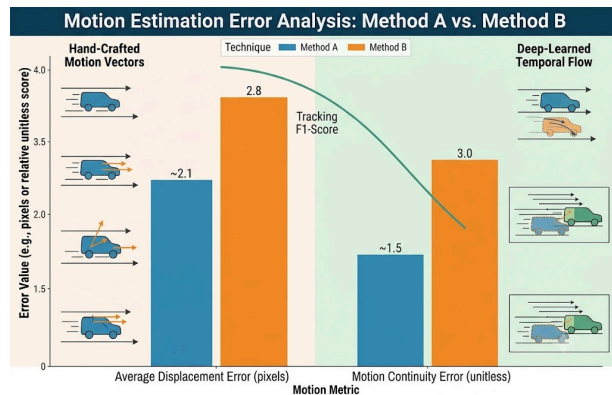
Method B

$$\text{Average Error}_B = \frac{3 + 5 + 4}{3} = \frac{12}{3} = \boxed{4 \text{ pixels}}$$

(b) Comparison

Method	Displacement Errors (pixels)	Average Error
Method A	2, 3, 2	2.33 pixels
Method B	3, 5, 4	4 pixels

Since lower displacement error indicates more accurate motion estimation, Method A provides better accuracy.



3. Depth Estimation Comparison

Given Data

Object	Actual Depth (m)	Stereo Vision (m)	SfM (m)
Object 1	2.0	2.0	2.5
Object 2	3.0	3.0	3.8
Object 3	4.0	4.0	5.0

(a) Absolute Error for Each Object

Formula:

Absolute Error=| Estimated Depth–Actual Depth |

Stereo Vision

Object 1:

$$| 2.0-2.0 | =0 \text{ m}$$

Object 2:

$$| 3.0-3.0 | =0 \text{ m}$$

Object 3:

$$| 4.0-4.0 | =0 \text{ m}$$

Therefore:

Stereo Vision errors=0, 0, 0 m

Structure-from-Motion (SfM)

Object 1:

$$|2.5 - 2.0| = 0.5 \text{ m}$$

Object 2:

$$|3.8 - 3.0| = 0.8 \text{ m}$$

Object 3:

$$|5.0 - 4.0| = 1.0 \text{ m}$$

Therefore:

SfM errors=0.5, 0.8, 1.0 m

(b) Average Absolute Error

Stereo Vision

$$\text{Average Error} = \frac{0 + 0 + 0}{3} = 0 \text{ m}$$

SfM

$$\text{Average Error} = \frac{0.5 + 0.8 + 1.0}{3} = \frac{2.3}{3} = 0.767 \text{ m}$$

(c) Comparison

Method	Object 1 Error	Object 2 Error	Object 3 Error	Average Error
Stereo Vision	0 m	0 m	0 m	0 m
SfM	0.5 m	0.8 m	1.0 m	0.767 m

SfM has an average absolute error of approximately 0.767 m, with errors increasing from 0.5 m to 1.0 m.

Stereo Vision is more accurate than SfM

